

requirements are small size, high sound quality and reliability. It has wide applications in mobile terminals, speech recognition, portable media players, laptop and notebook computers, gaming and virtual reality (VR) input devices.

Acoustic sensor. It contains advanced microstructure technology to provide fast and sensitive responses to flow amount/direction of air. It is used in industrial, commercial and medical applications.

Torque sensor. This sensor is used for measuring and recording the torque for engines, crankshafts, gearboxes, rotors and more. It converts mechanical input into electrical output. There can be static torque or dynamic torque measurement.

Vibration sensor. It measures motor vibration and helps in shock analysis for automobiles and aircraft. There can be impulse or continuous vibration measurements. This sensor measures the effect of vibration on an object under real operating conditions.

Current sensor. This sensor provides fast and accurate responses for power measurement. It measures the current-flowing capabilities of the device under different operating conditions.

Force sensor. It measures force with proportional output, along with interface pressures between two mating surfaces. It is used in the automotive industry to assist engineers with T&M, R&D and quality-control applications.

Humidity sensor. Humidity measurement is critical for the food processing industry, for the assurance of quality from production to storage and transport to final users.

This type of sensor is also used in various types of air-conditioning units. It helps in energy consumption, too.

Inertial measurement unit. These units provide motion, position and navigation sensing from a single device over six-degrees-of-freedom.

Industrial applications

Industries served by sensors include electronic appliances, business machines, environmental testing, food and beverage, injection moulding,

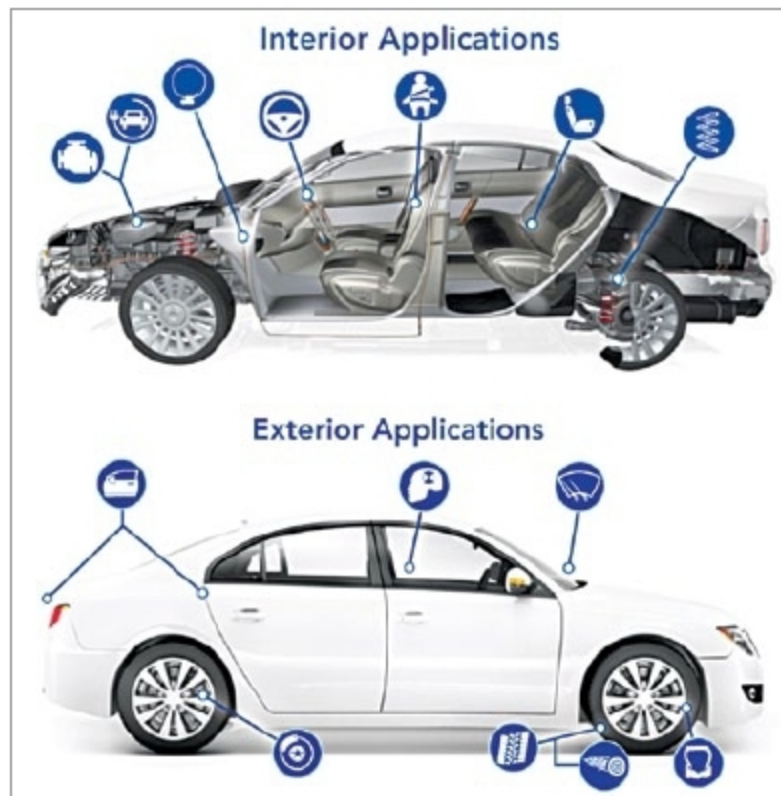


Fig. 2: Automotive applications of sensors
(Credit: www.tekscan.com)

medical, quality assurance, package design and testing, production and process equipment, pulp and paper, and smart homes, among others. Other critical application industries are ground and flight testing, automobiles, defence and military, energy and utilities, oil and gas, mines, sub-sea communications and railways.

Map interface provides key information such as total force, peak pressure and centre of force. For better understanding of surface behaviour design validation, it is important to verify the structural integrity of lighter and cost-saving components. Other factors include research and design, quality control and manufacturing tool assist in machine setup, identifying failure modes, vendor validation, reducing downtime and FEA validation.

Automotive. An automotive measurement system measures noise, vibration and harshness of component and system performance, like ride

handling and drivability, road load data acquisition in electric vehicles, powertrain development and off-highway drives. Sensor technology can assist with automotive seat comfort, measuring brake pad pressure distribution, tyre tread patterns, wiper blade force profiles and more.

Pressure measurement between a human body and support surface helps assess comfort, design and ergonomics.

Nip measurement verifies uniform pressure distribution between rollers by capturing nip pressure profile data in real time and windshield wiper. To improve system performance, it is important to evaluate pressure distribution and wiper blade force profiles.

Aerospace and defence. A T&M system provides testing for ground, flight, explosives, environment, guns and their impact, underwater sensors and satellites. Specific applications can be handled by sensors equipped with software and electronics, like grip for robots, and measuring and evaluating static and dynamic pressures while grasping objects.

To sum up

A sensor mapping system consists of data-acquisition electronics, sensors and software, which run directly on a laptop or desktop PC. To upgrade a system, various sensor models and software are available. Impossible insights have been made possible through minimally-invasive force and pressure measurements.

Measurement and test sensors are electronic transducers designed for use in test equipment in R&D or for individual users. These can be permanently installed on industrial and process systems in applications such as Industrial Internet of Things (IIoT). There are measurement and pressure transmitters for flow, pressure, temperature, rotations per minute, current, voltage and other measurable factors. **EFY**

Manufacturers of high-end sensors for T&M

- AI-Tek Instruments
- HBM India
- Honeywell International Inc.
- PCB Piezotronics Inc.
- Sensit Inc.
- STMicroelectronics
- TE Connectivity Ltd